

A Project Report

SMART HOME AUTOMATION SYSTEM: IoT-BASED HOME SAFETY AND AUTOMATION USING ESP8266



CSE441: Internet of Things

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Abstract

Modern households increasingly rely on manual switches and constant human presence to stay safe and comfortable, which leaves them vulnerable to hazards such as undetected fire, unmonitored intrusions, and inconvenient appliance control. This project, titled “Smart Home Automation System,” presents an affordable Internet of Things (IoT) based solution for home safety and automation built around an **ESP8266 microcontroller**.

The system continuously monitors the home environment using a flame sensor for fire detection, a PIR sensor for outdoor motion sensing, and a **DHT11 sensor** for temperature and humidity readings. When fire is detected, the **ESP8266** immediately triggers a buzzer alarm at the location and pushes a real-time alert to a web dashboard. All sensor data is synchronized with a cloud database (Firebase) so that the homeowner can view live readings, motion status, and fire alerts from any internet-connected device.

In addition to monitoring, the system allows remote control of a room light and a water pump through relay modules, which can be switched on or off either from the web dashboard or manually through push switches for local override. The web dashboard is designed to be simple and accessible, presenting live temperature and humidity values, a fire-alert banner, motion status, and ON/OFF controls for both appliances.

The proposed system was evaluated based on the reliability of fire detection, the responsiveness of remote-control commands, and the accuracy of environmental readings. The project demonstrates that a small set of affordable, readily available components can be combined into a practical smart-home solution that improves safety and convenience. Planned future enhancements include a dedicated mobile application, voice control through assistants such as Google Assistant or Alexa, solar-powered backup, and AI-based prediction of fire risk.

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Chapter 1

INTRODUCTION

1.1 Problem Specification / Statement

Traditional homes depend heavily on manual switches and constant human presence for both safety and comfort. Lights and water pumps must be operated in person, and there is no way to check on the house — its temperature, humidity, or any unusual activity outside — while the occupants are away. Fires, in particular, can spread rapidly, and without an early detection and alert mechanism, the risk to property and life increases sharply. This project addresses these gaps by developing a portable, low-cost IoT-based smart home system using an ESP8266 microcontroller that senses fire, motion, temperature, and humidity, and allows remote control of a light and a water pump through a web dashboard.

1.2 Objectives

- Develop an affordable, IoT-based home safety and automation system using an ESP8266 microcontroller.
- Detect fire early by sensing flame and triggering an instant buzzer alarm.
- Monitor the environment by tracking real-time temperature and humidity using a DHT11 sensor.
- Sense outside activity by detecting motion using a PIR sensor and displaying it live on the dashboard.
- Control appliances remotely, allowing the light and water pump to be turned on or off from a website.
- Provide a simple, accessible web dashboard that reflects sensor data and accepts control commands in real time.

1.3 Motivation and Scope

The motivation behind this project is to make homes safer and more convenient without requiring expensive or complicated equipment. Interpreters of danger — such as smoke or fire — are not always available to homeowners in the moment, and manual control of appliances is inconvenient

when residents are away. This project aims to close that gap with an affordable system that senses hazards, raises alerts immediately, and can be controlled remotely from any internet-connected device.

The scope of the project covers fire detection through a flame sensor, environmental monitoring through a DHT11 sensor, outdoor motion detection through a PIR sensor, and remote control of a light and a water pump through relay modules. Sensor data and control commands are synchronized through a Firebase cloud database, and presented through a web dashboard accessible from any device. Manual override through push switches is also provided for local, in-person control. Future enhancements, such as a dedicated mobile application, voice control, solar-powered backup, and AI-based fire-risk prediction, are discussed as extensions beyond the current scope.

1.4 Organization of Project Report

This project report is organized into the following chapters, each addressing a specific aspect of the proposed Smart Home Automation System:

- Chapter 1: Introduction — Presents the background, problem, objectives, scope, and motivation of the project.
- Chapter 2: Background — Discusses existing home-automation and safety systems, along with the tools and technologies used.
- Chapter 3: Implementation — Describes the front-end dashboard design, back-end logic, and system modules.
- Chapter 4: User Manual — Lists the hardware and software requirements and explains how to use the system.
- Chapter 5: Conclusion — Summarizes the project outcomes, its limitations, and possible future work.

Chapter 2

BACKGROUND

2.1 Existing System Analysis

A range of home-automation and safety systems already exist, built using varying combinations of sensors, microcontrollers, and connectivity options. Commercial smart-home hubs (such as Google Nest and Amazon Alexa-based ecosystems) offer voice control and app-based automation, but they typically rely on proprietary hardware and cloud services, making them relatively expensive for a basic safety-and-control use case.

Standalone fire-alarm systems are widely available and reliable for detecting smoke or heat, but most operate in isolation — they sound a local alarm without notifying the homeowner remotely or integrating with other home functions such as lighting or irrigation control. Similarly, many DIY IoT projects implement either environmental monitoring or remote appliance control individually, rather than combining safety detection, monitoring, and control into a single affordable dashboard.

Compared to these approaches, the proposed system integrates fire detection, temperature and humidity monitoring, motion sensing, and remote appliance control into one ESP8266-based platform with a shared web dashboard, aiming to offer broader functionality at a significantly lower cost than commercial alternatives.

2.2 Supporting Literature

This project draws on concepts from embedded systems, sensor interfacing, cloud databases, and web-based dashboards. The ESP8266 is a widely used, low-cost Wi-Fi-enabled microcontroller well suited for IoT applications that require sensor reading and network communication in a compact form factor. Flame sensors detect fire by sensing infrared light emitted by flames, while PIR (Passive Infrared) sensors detect motion by sensing changes in infrared radiation from moving objects such as people.

The DHT11 sensor is a common, low-cost digital sensor used for reading ambient temperature and humidity, and is frequently used in similar IoT monitoring projects. Firebase, a cloud platform offering a real-time database, is used to synchronize sensor readings and control commands between the ESP8266 and the web dashboard with minimal latency. Relay modules allow the low-power microcontroller to safely switch higher-power devices, such as a light or a water pump, on and off. Together, these components and tools were selected for their affordability, wide community support, and proven reliability in real-time IoT applications.

Chapter 3

IMPLEMENTATION

3.1 Interface Design / Front-End

The front-end of the system is a simple, accessible web dashboard designed to present live information at a glance. It displays real-time temperature and humidity readings that update automatically, a fire-alert banner that appears instantly when the flame sensor detects fire, and a motion-status indicator showing whether someone is currently detected outside the house. The dashboard also provides ON/OFF buttons for the room light and the water pump, usable from any internet-connected device such as a phone, tablet, or laptop. The layout emphasizes clarity, ensuring that critical safety information — particularly fire alerts — is immediately visible.

3.2 Back-End

The back-end logic runs on the ESP8266 microcontroller, which continuously reads the connected sensors — the flame sensor, the PIR sensor, and the DHT11 sensor — and synchronizes their readings with a Firebase real-time cloud database over Wi-Fi. When a control command (for example, turning the light or pump on or off) is issued from the web dashboard, it is written to the Firebase database; the ESP8266 continuously checks the database for updates and switches the corresponding relay module accordingly. This architecture keeps the embedded firmware lightweight, offloading data storage and dashboard rendering to the cloud and browser respectively, while the ESP8266 focuses on sensing and actuation.

3.3 Modules

The system is organized into the following functional modules:

- Fire Detection Module — Uses the flame sensor to detect fire and immediately triggers a local buzzer alarm and a dashboard alert.
- Environmental Monitoring Module — Uses the DHT11 sensor to track and report real-time temperature and humidity.
- Motion Detection Module — Uses the PIR sensor to detect outdoor motion and reflect it live on the dashboard.
- Light Control Module — Allows the room light to be switched on or off remotely via a relay, or locally via a push switch.

- Water Pump Control Module — Allows the water pump to be started or stopped remotely via a relay, or locally via a push switch.
- Cloud Synchronization Module — Synchronizes sensor data and control commands between the ESP8266 and the web dashboard through Firebase.

Chapter 4

USER MANUAL

4.1 System Requirement

4.1.1 Hardware Requirement

The hardware required for this project includes:

1. ESP8266 Board
2. Flame Sensor
3. PIR Sensor
4. Humidity and Temperature Sensor (DHT11)
5. 2 Push Switches
6. 2 Relay Modules
7. Water Pump
8. LED (6V)
9. Buzzer
10. 9V Battery
11. Wi-Fi Network

4.1.2 Software Requirement

1. Arduino IDE
2. ESP8266 Board Package
3. Firebase Real-Time Database
4. Web Browser (for dashboard access)
5. HTML / CSS / JavaScript (dashboard front-end)

4.2 User Interfaces

The system exposes its interface through a single web dashboard, accessible from any internet-connected device. The main sections of the dashboard are described below.

1. Live Environment Panel — Displays real-time temperature and humidity readings, updated automatically as new sensor data arrives.
2. Fire Alert Banner — Appears instantly across the dashboard when the flame sensor detects fire, alongside the local buzzer alarm.
3. Motion Status Panel — Shows whether outdoor motion is currently detected by the PIR sensor.
4. Appliance Control Panel — Provides ON/OFF buttons for the room light and the water pump, with the current status reflected on-screen.

No login credentials are required to view the current version of the dashboard, as it is intended for private, single-household use over a local Wi-Fi network; access control may be added in a future revision.

Chapter 5

CONCLUSION

5.1 Conclusion

This project has demonstrated that a small set of affordable, readily available components — an ESP8266 microcontroller, a flame sensor, a PIR sensor, a DHT11 sensor, and relay-driven actuators — can be combined into a practical smart-home system that genuinely improves safety and convenience. By integrating fire detection, environmental monitoring, motion sensing, and remote appliance control into a single web dashboard, the project delivers a low-cost solution suitable for everyday households. Beyond the technical outcome, the project gave the team hands-on experience in embedded systems, sensor interfacing, cloud synchronization, and real-time web control.

5.2 Limitations

The current system has a few practical limitations. It depends on a stable Wi-Fi connection for cloud synchronization and remote control, meaning it cannot operate independently if the network is unavailable. Sensor accuracy — particularly for the flame and PIR sensors — can be affected by environmental factors such as lighting conditions, distance, and placement. The system currently supports only one light and one water pump; scaling to additional appliances would require further relay channels and dashboard updates. There is also no user authentication on the dashboard in its current form, which would need to be addressed before wider deployment.

5.3 Future Works

- Develop a dedicated mobile application for Android and iOS for more convenient control and notifications.
- Add voice control support using assistants such as Google Assistant or Amazon Alexa.
- Introduce solar-powered backup to keep the system operational during power outages.
- Incorporate AI-based prediction to flag fire risk before it fully develops, using sensor trends over time.
- Add user authentication to the web dashboard to restrict access to authorized household members.
- Extend the system to control additional appliances beyond the light and water pump.

References

1. "ESP8266 Overview." Espressif Systems, www.espressif.com/en/products/socs/esp8266.
2. "Firebase Realtime Database." Google Firebase, firebase.google.com/docs/database.

Appendix A